Arab academy for Science, Technology and Maritime Transport

Binary Multiplication

Rules of Binary Multiplication

- **0 × 0 = 0**
- **0 × 1 = 0**
- **1 × 0 = 0**
- **1 × 1 = 1**

Example:

1) $(1011)_2 \times (101)_2$

$$
\begin{array}{r}
1\ 0\ 1\ 1 \\
\times\ \ 1\ 0\ 1 \\
\hline
1\ 0\ 1\ 1 \\
1\ 0\ 1\ 1\ \ \ \\
+\ 0\ 0\ 0\ 0\ \ \\
+\ 1\ 0\ 1\ 1\ \ \ \\
\hline
1\ 1\ 0\ 1\ 1\ 1
\end{array}
$$

2) $(1101)_2 \times (11)_2$

$$
\begin{array}{r}
1\ 1\ 0\ 1 \\
\times\ \ \ 1\ 1 \\
\hline
1\ 1\ 0\ 1 \\
+\ 1\ 1\ 0\ 1\ \ \ \\
\hline
1\ 0\ 0\ 1\ 1\ 1
\end{array}
$$

3) $(1010)_2 \times (111)_2$

$$
\begin{array}{r}
1\ 0\ 1\ 0 \\
\times\ \ 1\ 1\ 1 \\
\hline
1\ 0\ 1\ 0 \\
1\ 0\ 1\ 0\ \ \ \\
+\ 1\ 0\ 1\ 0\ \ \ \\
\hline
1\ 0\ 0\ 0\ 1\ 1\ 0
\end{array}
$$

Multiply the following binary numbers:

1. $(101)_2 \times (11)_2$
2. $(1101)_2 \times (10)_2$
3. $(1010)_2 \times (101)_2$
4. $(111)_2 \times (111)_2$
5. $(1001)_2 \times (110)_2$
6. $(1100)_2 \times (1011)_2$
7. $(10010)_2 \times (101)_2$

Multiply the following decimal numbers using 8-bit Two's Complement binary representation:

1. $5 \times (-3)$
2. $7 \times (-5)$
3. -2×5
4. $13 \times (-9)$
5. -17×2
6. $-8 \times (-4)$

Reminder: Represent both operands using 8-bit Two's Complement. Multiply and express each answer as an 8-bit signed binary number. (e.g., 5 = 00000101; −2 = 11111110).